5. Human Machine Interface

5.1. Egg Geometry

Before any subsystem calculation can be performed, a consistent geometric model of the egg must be established. This model is referenced throughout all subsequent sections.

5.1.1. Volume

The target egg is a standard large egg of 60g. A fresh egg has a density in the range 1.03 - 1.10g/mL (midpoint 1.065g/mL), giving a volume of approximately 56.3cm3. Since the density is close to 1g/cm3, the volume is taken directly as 60cm3. This is a conservative upward rounding that slightly overestimates surface area and content volume, resulting in conservative outputs in the heat transfer section.

5.1.2. Sphere Model

V_{egg} = 60 cm^3 (simplified) The egg is modelled as a perfect sphere of equivalent volume. This simplifies all downstream area and geometry calculations.

$$r = \sqrt[3]{\frac{3V}{4\pi}} = \sqrt[3]{\frac{3 \times 60}{4\pi}} = \sqrt[3]{14.32} = 2.426 \text{ cm}$$

$$d_{\text{egg}} = 2r = 4.85 \text{ cm}$$

$$A_{\text{sphere}} = 4\pi r^2 = 4\pi(2.426)^2 = 73.93 \text{ cm}^2$$

5.1.3. Shape Correction Factor

A real egg is a prolate spheroid. Using a typical axis ratio a/b = 1.25 and the same volume, the true surface area is computed and compared to the sphere model.

$$V = \frac{5}{3}\pi b^3 = 60 \text{ cm}^3 \quad \Longrightarrow b = 2.256 \text{ cm}, \qquad a = 2.820 \text{ cm}$$

With eccentricity $e = \sqrt{1 - \left(\frac{b}{a}\right)^2} = 0.600$

$$A_{\text{spheroid}} = 2\pi b^2 \left(1 + \frac{a}{be} \arcsin e\right) = 74.86 \text{ cm}^2$$

$$k_{\text{shape}} = \frac{A_{\text{spheroid}}}{A_{\text{sphere}}} = \frac{74.86}{73.93} = 1.013$$

The sphere model introduces a surface area error of 1.3%. The correction factor k_{shape}= 1.013 is applied to all surface area terms in the heat transfer section.

5.1.4. Geometry Summary

Parameter	Value	Note
Egg mass	60 g	Standard large egg
Egg volume (simplified)	60 cm^3	Conservative upward rounding
Sphere diameter	4.85 cm	
Sphere radius	2.426 cm	
Surface area (sphere)	73.93 cm^2	
Shape correction k	1.013	Applied to surface area only
Shell mass (10%)	6 g	
Content mass	54 g	$m_{egg} - m_{shell}$

Table 15 : Geometric Summary of our egg dimensions.

5.2. Process Time

This section estimates the total preparation time before the egg enters the cooking stage. The preparation time includes oil pre-spray, gate release, egg transport, breaking mechanism actuation, egg draining, and spring reset. The calculated preparation time is then compared with the conveyor timing to verify whether the 30 s cooking-cycle target can be achieved.

5.2.1. Stage 0 - Oil Pre-Spray

The oil pre-spray stage prepares the cooking plate before the egg is released. This stage includes the MOSFET switching delay, pump rotor spin-up, oil delivery through the tube, check valve closure, and oil spreading over the heated plate.

Assumptions:
• MOSFET: IRLZ44N.
• Gate resistance: 10 Ω.
• Effective gate capacitance: 1.8 nF.
• Pump rotor mass: 5 g.
• Pump rotor radius: 8 mm.
• Pump torque: 0.005 Nm.
• Required oil volume: 2 mL.
• Pump flow rate: 50 mL/min.
• Oil tubing is pre-filled during system initialization.
• Check valve closure time: 1.0 ms.
• Estimated oil spreading time: 400.0 ms.
• Plate diameter: 15 cm.

a.MOSFET gate charge (IRLZ44N)

The MOSFET switching time is extremely small compared with the mechanical and fluid delivery times, so it is treated as approximately 0.00 ms in the total timing calculation.

$$\tau = RC = 10\ \Omega \times 1.8\ \mathrm{nF} = 18\ \mathrm{ns}$$
$$t_{\mathrm{MOSFET}} = 2.2\tau = 2.2(18\ \mathrm{ns}) = 39.6\ \mathrm{ns} \approx 0.00\ \mathrm{ms}$$

b.Pump rotor spin-up

The pump rotor spin-up time is calculated from the rotor moment of inertia and angular acceleration:

$$J = \frac{1}{2}mr^2 = \frac{1}{2}(0.005)(0.008)^2 = 1.6 \times 10^{-7}\ \mathrm{kg\,m^2}$$
$$J = 0.16\ \mu\mathrm{kg\,m^2}$$

$$\alpha = \frac{T}{J} = \frac{0.005}{1.6 \times 10^{-7}} = 31\,250 \text{ rad/s}^2$$

$$t_{\text{spin-up}} = \frac{\omega}{\alpha} = \frac{100}{31\,250} = 0.00320 \text{ s} = 3.20 \text{ ms}$$

c.Oil tubing travel (pre-filled at boot)

Since the oil tube is assumed to be pre-filled during system initialization, the oil tubing travel delay is taken as:

$t_{\text{tube}} = 0.00 \text{ ms}$

d.2 mL delivery @ 50mL/min

The oil delivery time is calculated from the required oil volume and the pump flow rate:

$$t_{\text{delivery}} = \frac{V}{Q} = \frac{2.0}{50/60} = 2.400 \text{ s} = 2400.0 \text{ ms}$$

e.Check valve closure (spring loaded)

After delivery, the spring-loaded check valve closes. This delay is estimated as:

$t_{\text{valve}} = 1.0 \text{ ms}$

f.Oil spreading on the plate

The oil film thickness over a 15 cm diameter plate is estimated from the delivered oil volume and the plate surface area:

$$\delta_{\text{film}} = \frac{V}{A} = \frac{2 \text{ mL}}{\pi(7.5 \text{ cm})^2}$$

$$\delta_{\text{film}} = \frac{2 \text{ cm}^3}{176.7 \text{ cm}^2} = 0.0113 \text{ cm} = 113 \ \mu\text{m}$$

At the heated plate temperature, the oil viscosity decreases significantly. The spreading delay is therefore estimated as:

$t_{\text{spread}} = 400.0 \text{ ms}$

The total oil pre-spray time is:

$$t_0 = t_{\text{MOSFET}} + t_{\text{spin-up}} + t_{\text{tube}} + t_{\text{delivery}} + t_{\text{valve}} + t_{\text{spread}}$$

$$t_0 = 0.00 + 3.20 + 0.00 + 2400.0 + 1.0 + 400.0$$

$$t_0 = 2804.20 \text{ ms} = 2.804 \text{ s}$$

Therefore, the oil pre-spray stage requires approximately 2804.20 ms, or 2.804 s. This stage is dominated by the oil delivery time, while the MOSFET switching delay is negligible.

5.2.2. Stage 1 - Gate Release

The gate-release stage begins when the first IR sensor detects that an egg is present at the release position. The sensor signal is read by the STM32 microcontroller, which then activates the solenoid driver. The solenoid retracts the gate and allows one egg to enter the transport channel.

Assumptions:
• IR sensor: GP2Y0A21.
• Sensor sampling frequency: 2 kHz.
• STM32 clock frequency: 64 MHz.
• EXTI interrupt latency: 3 clock cycles.
• GPIO switching instruction count: approximately 12 instructions.
• MOSFET gate resistance: 10 Ω.
• MOSFET effective gate capacitance: 1.8 nF.
• Solenoid coil inductance: 10 mH.
• Solenoid coil resistance: 5 Ω.
• Solenoid plunger begins moving at approximately 50% of maximum current.
• Gate solenoid stroke length: 10 mm.
• Solenoid force: 10 N.
• Opposing spring force: 0.5 N.
• Moving plunger mass: 15 g.
• Egg release lag due to inertia: 1.000 ms.

a.IR Sensor 1 (GP2Y0A21, 2 kHz)

$$t_{\text{sensor}} = \frac{1}{f} = \frac{1}{2000} = 0.0005 \text{ s} = 0.500 \text{ ms}$$

b.STM32 EXTI latency @ 64 MHz

The STM32 external interrupt latency is calculated from the controller clock frequency:

$$t_{\text{EXTI}} = \frac{3}{64 \times 10^6} = 46.9 \text{ ns} \approx 0.000 \text{ ms}$$

c.ISR + HAL_GPIO_WritePin

The time required for the interrupt service routine and GPIO output command is:

$$t_{\text{ISR}} = \frac{12}{64 \times 10^6} = 187.5 \text{ ns} \approx 0.000 \text{ ms}$$

d. MOSFET gate charge (solenoid driver)

The MOSFET gate charging time for the solenoid driver is:

$\tau = RC = 10\ \Omega \times 1.8\ \text{nF} = 18\ \text{ns}$

$t_{\text{MOSFET}} = 2.2\tau = 2.2(18\ \text{ns}) = 39.6\ \text{ns} \approx 0.000\ \text{ms}$

e. Solenoid coil energisation

The solenoid coil energisation time is calculated from the electrical time constant of the coil. The plunger is assumed to begin moving when the current reaches approximately 50% of its maximum value:

$\tau = \frac{L}{R} = \frac{10\ \text{mH}}{5\ \Omega} = 2.0\ \text{ms}$

$t_{\text{coil}} = -\tau \ln(1 - 0.5) = -2.0 \ln(0.5) = 1.386\ \text{ms}$

f. Gate solenoid mechanical stroke 10 mm

The solenoid mechanical stroke time is calculated using constant-acceleration motion:

$F_{\text{net}} = F_{\text{solenoid}} - F_{\text{spring}} = 10 - 0.5 = 9.5\ \text{N}$

$a = \frac{F_{\text{net}}}{m} = \frac{9.5}{0.015} = 633.3\ \text{m/s}^2$

$t_{\text{stroke}} = \sqrt{\frac{2L}{a}} = \sqrt{\frac{2(0.010)}{633.3}} = 0.005620\ \text{s} = 5.620\ \text{ms}$

g. Egg release lag (inertia response)

The egg release lag is estimated as:

$t_{\text{egg lag}} = 1.000\ \text{ms}$

The total gate-release time is:

$t_1 = t_{\text{sensor}} + t_{\text{EXTI}} + t_{\text{ISR}} + t_{\text{MOSFET}} + t_{\text{coil}} + t_{\text{stroke}} + t_{\text{egg lag}}$

$t_1 = 0.500 + 0.000 + 0.000 + 0.000 + 1.386 + 5.620 + 1.000$

$t_1 = 8.506\ \text{ms}$

Therefore, the gate-release stage requires approximately 8.506 ms. The dominant part of this stage is the mechanical solenoid stroke, while the microcontroller and MOSFET switching delays are negligible compared with the mechanical motion.

5.2.3. Stage 2 - Egg Transport

The egg transport stage begins after the release gate opens. The egg rolls through the inclined channel, passes through a curved fillet transition, settles into the aluminium holder, and is then detected by the second IR sensor before the breaking mechanism starts.

Assumptions:
• Egg mass: 60 g.
• Egg radius used for rolling model: 28 mm.
• Egg is approximated as a solid sphere for rolling inertia.
• Main transport channel angle: 30°.
• Main channel length: 20 cm.
• Kinetic friction coefficient on PP surface: 0.12.
• Fillet transition radius: 6 cm.
• Fillet arc angle: 30°.
• Velocity loss through fillet transition: 1.5%.
• Holder settling angle: 15°.
• Holder settling length: 5 cm.
• Holder friction coefficient: 0.15.
• Impact damping time in rubber-padded holder: 15.00 ms.
• IR Sensor 2, STM32, and MOSFET signal chain delay: 1.887 ms.

a.Egg moment of inertia

$$I = \frac{2}{5}mR^2 = \frac{2}{5}(0.060)(0.028)^2$$

$$I = 18.82 \times 10^{-6}\ \mathrm{kg\,m^2}$$

$$K = 1 + \frac{I}{mR^2} = 1.400$$

b.Rolling check

The rolling condition is checked for the 30° PP transport surface:

$$\tan(30^\circ) = 0.577$$

$$\frac{\mu_s}{1 + \frac{mR^2}{I}} = 0.043$$

$$0.577 > 0.043$$

Therefore, the egg is expected to roll rather than slide on the inclined PP surface.

c.Channel roll (θ=30°, μ_k=0.12, L=20cm)

The acceleration down the 30° channel is:

$$a = \frac{g(\sin\theta - \mu_k \cos\theta)}{K}$$

$$a = \frac{9.81(\sin 30° - 0.12\cos 30°)}{1.400}$$

$$a = \frac{9.81(0.500 - 0.104)}{1.400} = 2.775 \text{ m/s}^2$$

The time required to roll through the 20 cm channel is:

$$t_{\text{channel}} = \sqrt{\frac{2L}{a}} = \sqrt{\frac{2(0.20)}{2.775}}$$

$$t_{\text{channel}} = 0.37964 \text{ s} = 379.64 \text{ ms}$$

$$v_{\text{exit}} = at = 2.775(0.37964) = 1.054 \text{ m/s}$$

d.Fillet transition (R=6cm, 30° arc)

The egg then passes through the curved fillet transition. The arc length is:

$$s_{\text{arc}} = R\theta = 0.06\left(\frac{\pi}{6}\right) = 0.03142 \text{ m} = 31.42 \text{ mm}$$

The normal force through the fillet is estimated as:

$$N_{\text{fillet}} = m\left(g\cos 30° + \frac{v^2}{R}\right)$$

$$N_{\text{fillet}} = 0.060\left(9.81\cos 30° + \frac{1.054^2}{0.06}\right) = 1.62 \text{ N}$$

With an estimated 1.5% velocity loss through the fillet, the average velocity is taken as:

$$v_{\text{avg}} = 1.038 \text{ m/s}$$

$$t_{\text{fillet}} = \frac{s_{\text{arc}}}{v_{\text{avg}}} = \frac{0.03142}{1.038} = 0.03027 \text{ s} = 30.27 \text{ ms}$$

e.Settle into Al holder (θ=15°, μ=0.15, L=5cm)

The egg then settles into the aluminium holder. The rolling condition at the 15° holder angle is:

$$\tan(15°) = 0.268$$

$$0.268 > 0.043$$

Therefore, the egg is still expected to roll during the settling phase.

The settling acceleration is:

$$a = \frac{g(\sin 15^\circ - 0.15 \cos 15^\circ)}{1.400}$$

$$a = 0.798 \text{ m/s}^2$$

Using the entry velocity into the holder, the estimated settling time is:

$$v_{\text{entry}} = 1.038 \text{ m/s}$$

$$t_{\text{settle}} = \frac{v_0 - v_f}{a} = 353.92 \text{ ms}$$

f.Impact damping in Al holder

The impact damping time in the rubber-padded aluminium holder is:

$$t_{\text{damping}} = 15.00 \text{ ms}$$

g.IR Sensor 2 + STM32 + MOSFET chain

The sensor and control signal delay after the egg reaches the second IR sensor is:

$$t_{\text{IR2+control}} = 1.887 \text{ ms}$$

The total egg transport time is:

$$t_2 = t_{\text{channel}} + t_{\text{fillet}} + t_{\text{settle}} + t_{\text{damping}} + t_{\text{IR2+control}}$$

$$t_2 = 379.64 + 30.27 + 353.92 + 15.00 + 1.887$$

$$t_2 = 780.72 \text{ ms}$$

5.2.5. Stage 4 - Drain and Spring Reset

The drain and spring reset stage begins after the egg shell has opened and the egg contents are released from the breaking holder. This stage includes the draining of the egg white and yolk through the outlet, the surface-tension drip delay, and the spring-return motion of the spear and wedge mechanisms.

Assumptions:
• Egg white volume: 33 mL.
• Egg yolk volume: 13 mL.
• Total drained egg content volume: 46 mL.
• Egg white density: 1040 kg/m³.
• Egg yolk density: 1030 kg/m³.
• Effective drain height: 35 mm.
• Egg white dynamic viscosity: 200 mPa·s.
• Discharge coefficient after viscosity correction: 0.597.
• Outlet radius: 15 mm.
• Surface tension: 48 mN/m.
• Spear return spring stiffness: 365,741 N/m.
• Spear moving mass: 105 g.
• Wedge return spring stiffness: 200,000 N/m.
• Wedge moving mass: 60 g.

a.Volume of whites and yolks.

$$V_{\text{total}} = V_{\text{white}} + V_{\text{yolk}} = 33 + 13 = 46 \text{ mL}$$

b.Torricelli drain (viscosity-corrected)

The ideal drain velocity is estimated using Torricelli's law:

$$v_{\text{ideal}} = \sqrt{2gh} = \sqrt{2(9.81)(0.035)} = 0.829 \text{ m/s}$$

Because egg white is viscous, the ideal velocity is reduced using a discharge coefficient:

$$v_{\text{drain}} = C_d v_{\text{ideal}} = 0.597(0.829) = 0.495 \text{ m/s}$$

The outlet area is:

$$A_{\text{out}} = \pi r^2 = \pi(15 \text{ mm})^2 = 706.9 \text{ mm}^2$$

The drain time is calculated from the drained volume, outlet area, and corrected drain velocity:

$$t_{\text{drain}} = \frac{V}{A_{\text{out}} v_{\text{drain}}}$$

$$t_{\text{drain}} = \frac{46 \text{ mL}}{706.9 \text{ mm}^2 \times 0.495 \text{ m/s}} = 131.6 \text{ ms}$$

c.Surface tension drip delay

After the main volume has drained, a short surface-tension drip delay is included:

$$t_{\text{drip}} = 3\sqrt{\frac{\gamma}{\rho g d}}$$

$$t_{\text{drip}} = 37.6 \text{ ms}$$

d. Spear return spring

The spear return motion is modelled as a spring-mass return system:

$$\omega_{\text{spear}} = \sqrt{\frac{k}{m}} = \sqrt{\frac{365741}{0.105}} = 1866 \text{ rad/s}$$

$$t_{\text{spear}} = \frac{\pi}{\omega_{\text{spear}}} = \frac{\pi}{1866} = 1.683 \text{ ms}$$

e. Wedge return spring

The wedge return motion is calculated in the same way:

$$\omega_{\text{wedge}} = \sqrt{\frac{k}{m}} = \sqrt{\frac{200000}{0.060}} = 1826 \text{ rad/s}$$

$$t_{\text{wedge}} = \frac{\pi}{\omega_{\text{wedge}}} = \frac{\pi}{1826} = 1.721 \text{ ms}$$

The total drain and spring reset time is:

$$t_4 = t_{\text{drain}} + t_{\text{drip}} + t_{\text{spear}} + t_{\text{wedge}}$$

$$t_4 = 131.6 + 37.6 + 1.683 + 1.721$$

$$t_4 = 172.57 \text{ ms}$$

Time for processing and prep stage grand total is divided as follows:

Stage 0 — Oil pre-spray	2804.20 ms	72.1%
Stage 1 — Gate release	8.51 ms	0.2%
Stage 2 — Egg transport	780.72 ms	20.1%
Stage 3 — Breaking	122.22 ms	3.1%
Stage 4 — Drain + reset	172.57 ms	4.4%

t_prep = 3888.21 ms = 3.888 s

5.3. Cooking Time

5.3.1. Core Assumptions

- Egg mass: m = 60 g = 0.060 kg
- Egg volume: V ≈ 60 cm³
- Egg properties (effective white + yolk):
- Thermal conductivity: k = 0.52 W/(m·K)
- Density: ρ = 1040 kg/m³
- Specific heat: cp = 3600 J/(kg·K)
- Thermal diffusivity: α = k/(ρ·cp) = 1.3889×10^{-7} m²/s
- Entry temperature from storage: T_init = 20°C

5.3.2. Geometry In Heat Direction

Half-thickness plate → yolk centre: L = 5.95 mm = 0.00595 m

1D finite difference grid:

- Nodes: N = 20
- Spatial step: dx = $\frac{L}{N} = \frac{0.00595}{20}$ = 0.2975 mm
- Time step: dt = 0.005 s
- Fourier number: Fo = α·dt/dx² ≈ 0.00785 < 0.5 (stable)

5.3.3. Boundary and Initial Conditions

Bottom plate (with thin oil film):

- Plate temperature: T_plate = 200°C
- Oil film: δ ≈ 113 µm, k_oil ≈ 0.17 W/(m·K) → h_eff ≈ 1500 W/(m²·K)
- Approximated as Dirichlet: T(0, t) = 200°C

Top IR heater:

- IR temperature: T_IR = 120°C
- Heat transfer coefficient: h_top = 20 W/(m²·K)
- Mixed (Robin) boundary at top node N:

$$T_N^{(n+1)} = \frac{\frac{k}{\Delta x} T_{N-1}^{(n+1)} + h_{\text{top}} T_{\text{IR}}}{\frac{k}{\Delta x} + h_{\text{top}}}$$

- Initial condition (20°C entry):
- T(x, 0) = 20°C (uniform)

- Targets:
- Whites set: centre temperature T_centre = 65°C → defines t_cook

- Yolk limit: centre temperature T_centre = 68°C → defines t_limit

5.3.4. 1D Finite Difference Scheme

- Interior nodes (i = 1..N-1):

$$T_i^(n+1) = T_i^n + Fo\cdot (T(i+1)^n - 2\cdot T_i^n + T(i-1)^n)$$

- Boundary nodes:
  - Bottom node 0: $T_0^(n+1) = 200°C$
  - Top node N: mixed condition as in section 3

5.3.5. Cook Time Results (20°C Entry , 200°C Cooking Plate)

- From the simulation:
  - Time to centre 65°C (whites set):
    t_cook = 23.305 s (≈ 23.30 s)

  - Time to centre 68°C (yolk limit):
    t_limit ≈ 25.0–25.5 s (approximate band)

  - Cook window:
    Δt_window ≈ t_limit − t_cook ≈ 1.7–2.0 s

5.3.6. Total Time (First Egg)

- Total first-egg time:
  T_total = t_prep + t_cook
  ≈ 3.888 s + 23.305 s
  ≈ 27.193 s

First egg ready in ≈ 27.2 s (centre at 65°C).

5.3.7. Conveyor & Plate Timing

- Heated conveyor length under plates:
  L_heated = 450 mm

- Conveyor speed for t_cook over 450 mm:
  v = L_heated / t_cook
  = $450/23.305$
  ≈ 19.3 mm/s

- Plate geometry:
  Plate length = 106 mm
  Gap between = 15 mm
  Centre spacing = d_center = 106 + 15 = 121 mm

- Time between plate centres at loading position:
  T_plate = d_center / v
  = 121 / 19.3
  ≈ 6.27 s

- Prep vs plate timing:
  t_prep ≈ 3.888 s
  T_plate ≈ 6.27 s
  Buffer = T_plate − t_prep
  ≈ 6.27 − 3.888
  ≈ 2.38 s

The next egg is fully prepped about 2.4 s before its plate arrives.

8. Electric Drive

8.1. Drive 1 - Conveyor (NEMA17 stepper)

8.1.1. Description of movement/task and estimated load

The conveyor drive moves two parallel 06B-1-K1 roller chains carrying four egg pocket plates around a closed loop (top straight 666.51 mm, bottom return 666.51 mm, two 180° wraps around 18T sprockets). Movement is unidirectional, indexed motion: each plate must dwell at the egg-breaking station, frying zone, and ejection point in turn. Target linear speed is 30 mm/s, giving ≈ 22.2 s travel over a single 666.51 mm straight.

Load m = 3.27 kg

The 3.27 kg total linear moving mass is taken from CAD mass properties in SolidWorks with AISI 316 stainless steel assigned to all custom parts, plus catalogued supplier mass for the 06B-1-K1 chain. It is the sum of all parts that translate with the chain:

Part	Qty	Mass each	Total
Egg pocket plate (AISI 316 sheet)	4	383.80 g	1.5352 kg
Egg payload (assumed 60 g per egg, max load)	4	60.00 g	0.2400 kg
Chain side assembly incl. connecting links (06B-1-K1)	2	724.84 g	1.4497 kg
SmCo magnets (Ø8 × 3 mm, ρ ≈ 8.3 g/cm³)	4	1.25 g	0.0050 kg
Plate-to-chain fastener sets (A4 M3 screw + nut + washers)	16	1.25 g	0.0200 kg
Retained food-grade lubricant film (allowance)	—	—	0.0200 kg
Total linear moving mass			**3.2699 kg ≈ 3.27 kg**

Table 23 : Conveyor System Load Split.

8.1.2. Summary of technical requirements

Requirement	Value
Motion type	Continuous unidirectional with indexed dwell
Target speed	30 mm/s (10.5 rpm at sprocket, 283.5 rpm at motor)
Speed range	0 → 30 mm/s (slow start, dwell-capable, no reverse needed)
Acceleration	0.030 m/s² (1 s ramp; very gentle to avoid egg slip)
Required output torque (with SF 2)	0.50 N·m at sprocket shaft
Required motor torque	≥ 25 mN·m at 283.5 rpm
Positioning accuracy needed	≤ 1 mm at pocket (alignment with breaker, fryer, ejector)
Resolution required	171.45 mm/rev / N_steps ≤ 1 mm → ≥ 172 steps/rev at output
Control mode	Position control (indexed open-loop stepping with home reference)
Homing	One-shot at power-up via Hall sensor on idler shaft (existing rotating-plate sensor approach reused)
Duty cycle	Continuous during cook cycle, intermittent overall
Ambient at motor location	≤ 50 °C (motor must sit outside heated zone — verified by 1.5 in input table)
Direction reversibility	Not required (one-way conveyor)

Table 24 : Conveyor System Technical Requirements.

8.1.3. Type of motor and reasons for specific choice

Choice: 2-phase hybrid stepper motor, open-loop position control.

Reasoning, in order of weight:

1. **Indexed positioning is the core requirement.** The conveyor must repeatably stop pockets under the egg breaker, then under the heating zone, then over the ejector. A stepper gives intrinsic, gearbox-multiplied positional resolution (200 × 27 = 5400 output steps/rev → 0.032 mm/step at the plate) with no encoder. A DC motor would need an encoder + closed-loop controller for the same job.
2. **Load is light and predictable.** Required motor torque (25 mN·m) is ~3 % of holding torque far inside the safe operating envelope where stepper torque loss at speed and missed-step risk are not concerns. There is no risk of stall under normal operation.
3. **Low speed is native territory for steppers.** 283.5 rpm is squarely in the high-torque band of a NEMA 17. A brushless DC motor would need significant gear reduction added on top of an already-coarse reduction; a brushed DC would suffer at this low speed with cogging and brush wear.
4. **Inertia ratio (0.31) is excellent,** well below the 5:1 comfort threshold for steppers, so acceleration/deceleration is crisp and resonance-free.
5. **Single-direction, no regen,** no need for a 4-quadrant drive. Stepper drivers are the cheapest solution for this profile.
6. **Project consistency.** The rotating plate subsystem already uses a stepper (28BYJ-48) and the same STM32 step/dir control firmware can be extended. One driver family across subsystems simplifies BOM, firmware, and diagnostics.

Why not alternatives:

- Brushed DC + encoder + PID: more firmware effort, encoder cost, and brush maintenance in a food/heat environment.
- BLDC + encoder: overkill for 25 mN·m at 283 rpm; controller cost and complexity not justified.
- Servo: same as BLDC; no advantage at this load and speed.
- AC induction with VFD: minimum practical motor sizes are far too large; positioning resolution is poor without an encode.

8.1.4. Motor choice and corresponding driver

Motor: NEMA 17 hybrid stepper — AM17HDB410-02N

Parameter	Value	Margin vs. requirement
Holding torque	0.85 N·m	34× the 25 mN·m needed
Phase current	1.8 A/phase	Sets driver minimum rating
Rotor inertia	123 g·cm²	Load ratio 0.31 — good
Shaft	Double (10 mm bore gearbox + free end available for optional encoder)	Future-proofs closed-loop upgrade
Mass	0.6 kg	Fixed mount, not relevant
Ambient rating	−20 to +50 °C	Requires mounting outside heated zone

Table 25 : NEMA 17 Motor Requirements.

Driver: TB6600 external stepper driver module.

Parameter	Value	Justification
Phase current	0.5 – 4.0 A peak, DIP-selectable	Set to 2.0 A peak (≈ 1.4 A RMS) for the 1.8 A/phase motor — full nameplate torque with thermal margin
Microstepping	1/1, 1/2, 1/4, 1/8, 1/16, 1/32 (DIP-selectable)	Use 1/8 → 1600 steps/rev × 27 = 43 200 steps/rev output → 0.004 mm/step at plate (resolution is not the binding constraint)
Supply voltage	9 – 42 V DC	Powered from the 24 V rail in your power tree
Logic inputs	STEP / DIR / EN, opto-isolated	3.3 V STM32 GPIO drives the opto LEDs through a series resistor (typical 270 Ω for 3.3 V logic)
Pulse frequency	Up to ~200 kHz	At 283.5 rpm motor × 1600 steps/rev = 7.56 kHz — well inside the linear range
Enclosure	Aluminium heatsink case, DIN-rail / panel mount	Mounts off-PCB on the control panel; survives ambient heat better than a chip-on-board solution

Table 26 : NEMA 17 Driver TB6600.

8.2. Drive 2 - Rotating Plate (28BYJ-48 stepper)

8.2.1. Description of Movement and Estimated Load

Movement & Task: The drive rotates a Ø250 mm stainless-steel plate that receives fried eggs from a dispenser located above a fixed dispensing point. The plate starts at a homed 0° position; egg 1 is deposited, then the plate indexes 90° and waits for egg 2, and so on for a total of four eggs spaced 90° apart around the plate, with 15 s between dispense events. Motion is purely rotational about the vertical axis, supported by a 4-inch (Ø102 mm) lazy-Susan bearing. Power is transmitted from the motor to the plate by a Ø38 mm rubber wheel (25.4 mm inner bore) coupled to the motor shaft and pressed against the outer edge of the plate (friction drive). A lever microswitch provides the absolute 0° home reference at start-up; an HX711 + load cell under the assembly detects the arrival of each egg.

Motion pattern: Three discrete index moves of 90° each (between eggs 1→2, 2→3, 3→4), separated by 15 s windows that include both the move and a dwell during which the next egg is dispensed and lands. After egg 4, the plate has rotated 270° from home; a final 90° (or reverse 270°) returns it to 0°.

Mass breakdown:

Rotating components :

- Stainless steel plate (Ø250 × 3 mm, 6061-T6 aluminium): **0.716 kg**
- Lazy-Susan bearing top race (rotating half of 92 g total for the bearing pair): ≈ 0.046 kg
- Rotating-side screws + centering ring (estimated half of the 0.2 kg total fastener allowance): ≈ 0.1 kg
- Dish (900 g) + 4 eggs ($4 \times 60g = 240g$): 1.14 kg (worst case, after the 4th egg lands)
- **Total rotating mass (worst case): ≈ 2.0 kg**

Weight on the bearing:

$$W = 2.0 \cdot 9.81 \approx 19.6N$$

Polar moment of inertia (worst case, treated as a uniform disk of r = 0.125 m):

$$J \approx \frac{1}{2} \cdot m \cdot r^2 = 0.5 \cdot 2.0 \cdot 0.125^2 \approx 0.016kg \cdot m^2$$

Load torque at the plate:

4-inch lazy-Susan friction (μ_eff ≈ 0.04 at race radius ≈ 0.040 m):

$$T_f \approx 0.04 \cdot 19.6 \cdot 0.040 \approx 0.031N \cdot m$$

Acceleration torque for a gentle ramp (0 → 1.5 RPM in 1 s, α ≈ 0.16 rad/s²):

T_a = J · α ≈ 0.016 · 0.16 ≈ 0.003 N·m (negligible)

Required torque at the plate (with margin): ~0.1 N·m

Transmission: The Ø38 mm wheel is coupled rigidly to the motor shaft via the coupler, they rotate 1:1. The Ø38 mm wheel driving the Ø250 mm plate edge is therefore a single-stage friction reduction of $i_friction = 250/38 \approx 6.58 : 1$ (speed down, torque up at the plate). One motor revolution rolls $pi \cdot 38 \approx 119mm$ of arc on the plate edge against the plate circumference $pi \cdot 250 \approx 785mm$. Referred to the motor shaft:

$T_motor \approx 0.1/6.58 \approx 0.015N \cdot m \approx 15mN \cdot m$

8.2.2. Summary of Technical Requirements

Requirement	Value
Rotating mass (worst case)	≈ 2.0 kg
Weight on bearing	≈ 19.6 N
Required torque at the plate (incl. margin)	≥ 0.1 N·m
Required torque at motor shaft (after 6.58:1 reduction)	≥ 15 mN·m
Motion profile	Indexed: 90° step, then dwell
Cycle time (move + dwell)	15 s per egg

Move duration / dwell split (chosen)	~10 s move, ~5 s dwell
Plate speed during move	~1.5 RPM (9 °/s)
Motor shaft speed during move	~9.9 RPM
Step rate to driver (half-step)	~673 Hz
Half-steps per 90° plate move	~6738
Total travel across 4 eggs	270° (0° → 90° → 180° → 270°)
Positioning accuracy	≤ 1° at the plate (dispense alignment)
Control type	Open-loop position control with absolute home reference (microswitch)
Holding during dwell	hold position against egg-drop impulse (~few N·s)
Direction	Bidirectional (forward indexing; reverse for re-home)
Duty cycle	Intermittent — short moves, short dwells, no continuous running

Supply	5 V DC (already available on the board as +5V_FUSED)
Logic interface	4 digital lines from MCU (PLATE_IN1…IN4)
Environment	Indoor, low vibration, no high temperature

Table 27 : Rotating Plate Technical Requirements.

8.2.3. Type of Motor - Choice and Reasoning

Stepper motor (open-loop). Gives discrete angular steps directly from a step count, holds position when energised, no feedback loop required, and matches the slow / low-torque / indexed-with-dwell profile of this task exactly.

Reasons:

- **Positioning without an encoder.** After homing on the microswitch at start-up, the MCU knows the absolute angle simply by counting steps. There's no back-drive path under normal use to cause step loss.
- **Native indexed-then-dwell behaviour.** A stepper holds its commanded position with the coils energised (and with significant detent torque even when de-energised, thanks to the internal gearbox). This is exactly what the task asks for: move 90°, stop, wait, repeat.
- **Speed range matches the task.** The plate only needs ~1.5 RPM during the move. Steppers are at their best in the low-speed region where DC motors would need heavy gearing.
- **Torque margin.** Required motor torque (~15 mN·m) is far below the holding torque of common small steppers, so there's no risk of stalling or losing steps under the egg-drop impulse.
- **Cost and integration.** A small unipolar stepper + 4-channel Darlington driver is the cheapest electronic drive that meets the requirements.

8.2.4. Motor and Driver Selection

Motor: 28BYJ-48, 5 V geared unipolar stepper.

Parameter	Value
Rated voltage	5 V DC

Phases	4 (unipolar, 5 wires)
Internal gearbox	~1 : 64
Step angle at output shaft (half-step)	$5.625^\circ/64 \approx 0.088^\circ \rightarrow$ 4096 half-steps/rev
Holding torque (output shaft)	≈ 0.3 N·m
Typical no-load speed	~15 RPM at output shaft
Current per phase	~150–200 mA at 5 V

Table 28 : 28BYJ-48 Motor Requirements.

Sanity check against the new geometry:

- **Torque:** $0.3N \cdot m \times 6.58 \approx \ 2.0N \cdot m$ **at the plate**, against ~0.1 N·m required → ~20× margin.
- **Speed:** 9.9 RPM required vs ~15 RPM available → ~1.5× margin (more comfortable than with the 8.33:1 reduction).
- **Resolution at the plate:** $0.088^\circ/6.58 \approx 0.013^\circ$ **per half-step**, far finer than the ≤1° target.
- **Load cell:** rotating + stationary above the load cell

  $\approx 2.0kg + \ 1.4kg = 3.4kg$ total. Within a 5 kg / ~49 N rated load cell with margin.

Driver: ULN2003A Darlington array, shown as U1 on the schematic.

- 7 open-collector Darlington channels rated 500 mA / 50 V each, comfortably above the ~200 mA per phase the 28BYJ-48 draws.
- Built-in flyback diodes to COM (tied to +5V_FUSED on pin 9) clamp the inductive kickback from the coils, which is essential for a unipolar stepper.
- Four inputs PLATE_IN1…IN4 from the MCU map directly to outputs O1…O4 driving PLATE_COIL1…COIL4 , The MCU outputs the step sequence (half-step pattern), the ULN2003A buffers and sinks the coil current.
- Only 4 of the 7 channels are used; the rest are tied off (I5…I7 crossed out, O5…O7 unused), matching the schematic.

8.3. Drive 3 - Oil System
8.3.1. Description of Movement and Estimated Load

Movement / task: The drive is a small peristaltic pump that delivers a controlled volume of cooking oil through a flat-fan nozzle onto each egg pocket on the rotating plate, before the egg is dispensed. There is **no positioned motion**, The pump simply needs to deliver a defined volume of fluid per dose. One dose = **2 mL**, and there are **four doses per customer** (one before each egg), giving **2-8 mL** per customer cycle.

Mechanism:

A peristaltic pump uses rollers on a rotor to compress flexible silicone tubing against a curved housing, advancing a slug of fluid with each rotation. The drive motor is a **small brushed DC gear motor** built into the pump head. Volume delivered is essentially the rotor's swept volume × revolutions, independent of back-pressure within the small range present here.

Hydraulic / mechanical load:

- **Oil:** cooking oil, ρ ≈ 920 kg/m³, viscosity ≈ 60–80 mPa·s at room temperature (falls quickly as it warms).
- Static head from tank to nozzle: a few tens of centimetres → only a few mbar of pressure drop. Negligible for the pump.
- **Tubing:** 4 × 6 mm food-grade silicone, short run (< 1 m). Pressure drop at the chosen flow rate is small, on the order of tens of mbar.
- **Nozzle:** 1/8", 65° flat-fan, flow rating ~0.1 (Spraying Systems "01" convention, ~0.4 L/min at 3 bar). At the very low pressures and flows used here, the nozzle is operating well below its rated point it will spray, but barely, which is exactly what's wanted to coat a small pocket.
- **Timing constraint (binding):** the oil must be in the pocket *before* the egg arrives. Egg-break to pocket landing $(t_prep) = 3.88s$, so the pump dispense + oil settle must fit inside that window with margin.
- **Required dispense volume per dose: 2 mL ± ~10%** (cooking oil dosing tolerates modest variation).
- **Required dispense time per dose: ~2.5 s** (leaves ~1.4 s of the 3.88 s window for oil flight, nozzle clearing, and settle in the pocket).
- **Required flow rate:** $2mL/2.5s = 0.8mL/s \approx 48mL/min.$ Comfortably inside the 30–100 mL/min range of a 12 V mini peristaltic pump.
- **Pump motor draw:** ~150–250 mA at 12 V (typical for this class), ~2–3 W input power.

8.3.2. Summary of Technical Requirements

Requirement	Value
Function	Volumetric dosing of cooking oil
Volume per dose	**2 mL**
Doses per customer cycle	**1- 4** (one before each egg)
Inter-dose interval	~15 s
Available window per dose (egg-break → pocket)	**3.88 s**
Dispense time per dose (chosen)	**~2.5 s**
Required flow rate	**~48 mL/min (≈0.8 mL/s)**
Margin within 3.88 s window	**~1.4 s** (oil settle + safety)
Volume repeatability	±10% acceptable
Control type	**Open-loop time-based dosing** (volume = flow rate × on-time)
Direction	Single-direction (no reverse needed)
Speed control	Optional PWM; simple on/off + timed pulse is sufficient
Back-flow prevention	Inline check valve keeps tubing primed between doses
Supply	12 V DC (already available on the board as +12V_FUSED)
Logic interface	1 digital line from MCU (PUMP_CTRL)
Duty cycle	Very intermittent — ~2.5 s on per dose, 1 - 4 doses spaced 15 s apart per customer, idle between customers

Environment	Inside machine enclosure; ambient ~30–45 °C; food-grade silicone in wetted path; no direct splash exposure

Table 29: Technical Requirements Table for Oil System

8.3.3. Type of Motor - Choice and Reasoning

From the morphological box, three oil-delivery options were carried forward to evaluation:

- **Gravity feed + solenoid valve.** Oil flows from an elevated tank through a solenoid valve when energised; dose volume = flow rate × open time. Electrically the simplest option (one solenoid, one MOSFET), and there's no motor at all. The problem is volumetric repeatability: flow rate depends on the head of oil in the tank (which drops as oil is used), on oil viscosity (which falls as the cooking surface warms the cabinet), and on any air bubbles in the line. Across a 600 mL tank the head changes by several centimetres, easily a 20–30% swing in flow, which exceeds our ±10% dose tolerance. The design also forces the tank to sit *above* the nozzle, which conflicts with the compact cabinet layout. Rejected on accuracy and packaging.
- **Diaphragm pump.** A reciprocating diaphragm driven by a small motor delivers oil in pulsed slugs. Self-priming, can handle higher pressures than a peristaltic, and tolerates dry running better. But the wetted path includes the diaphragm, inlet valve, and outlet valve all in contact with oil, all needing to be food-safe and serviceable. Small 12 V food-grade diaphragm pumps exist but cost noticeably more (~€15–25 vs ~€3), and the pulsed output makes precise 2 mL dosing harder than a continuous-flow peristaltic. The higher pressure capability is wasted here because we're spraying at near-zero back-pressure. Over-specified and more expensive without a corresponding benefit. Rejected.
- **Peristaltic pump.** Rollers on a rotor compress a flexible silicone tube against a curved housing, advancing a defined slug of fluid per revolution. Volume per second is set by motor speed and tube geometry, so timed open-loop control gives repeatable doses. **Chosen.**

Choice: Peristaltic pump with brushed DC gear motor, switched on for a calibrated dose time per egg.

Reasons:

1. **Repeatable volumetric dosing without feedback.** Per-revolution displacement is fixed by tube ID and rotor geometry, independent of tank level. At a constant supply voltage and constant on-time, dose-to-dose variation easily stays within the ±10% target something the gravity option cannot achieve.
2. **Food safety by construction.** Only the silicone tube touches the oil. The motor, gears, rotor, and rollers never contact the fluid path, and the tube is the single replaceable wear part. This is much simpler than the diaphragm pump, where multiple wetted parts (diaphragm + valves) all need to be food-grade and serviceable.
3. **No back-flow and no drip.** When the rollers stop, they occlude the tube and seal the fluid path. Combined with the inline check valve , there is no oil drip from the nozzle between doses a problem both gravity feed and a stopped diaphragm pump can suffer from.
4. **Self-priming, no air-lock issue.** The pump primes itself from an empty tube on first run and tolerates the occasional air bubble. Gravity feed by contrast becomes unpredictable in the presence of bubbles.
5. **Speed fits the 3.88 s prep window.** At full 12 V the pump delivers ~48 mL/min exactly enough to complete a 2 mL dose in ~2.5 s and leave ~1.4 s of settle time before the egg lands. The diaphragm pump's pulsed delivery would meet the average flow but make precise timing of the *end* of the dose less clean.
6. **Simple drive electronics.** A brushed DC motor switched on for a fixed time delivers the right dose one MOSFET as a low-side switch is all that's required. No H-bridge (no reverse), no microstepping driver, no encoder.
7. **Cost and packaging.** The complete 12 V mini peristaltic pump including motor, gearbox, rotor, rollers, and tube section is significantly cheaper than the diaphragm pump alternative, and only marginally more than the solenoid option (which would have lost on accuracy regardless of price).

8.3.4. Motor (Pump) and Driver Selection

Pump: 12 V mini Peristaltic pump , shown as Oil_Pump1 on the schematic.

Parameter	Typical value for this class
Rated voltage	12 V DC
Motor type	Brushed DC + built-in gearbox
Typical flow rate	30–100 mL/min (depends on tube ID and head speed)
Running current	~150–250 mA

Stall current	~400–500 mA
Tube fitting	4 mm ID silicone
Self-priming	Yes
Back-flow when idle	Negligible (rollers occlude the tube), reinforced by check valve

Table 30: Pump Selection Table

Driver: IRLZ44N logic-level N- channel MOSFET as a low-side switch (BOM #7), shown as Q_PUMP1 on the schematic.

Parameter	Value	Relevance
VDS	55 V	>> 12 V supply, large margin
ID continuous	47 A	>> 0.5 A pump stall current, vastly oversized
RDS(on) at VGS = 5 V	~22 mΩ	Logic-level — switches fully on from a 3.3 V or 5 V MCU pin
Power dissipation at 250 mA	$I^2 \cdot RDS(on) \approx 1.4mW$	Negligible — no heatsink required
Switching speed	Fast	Slowed deliberately by R4 = 100 Ω gate resistor to limit EMI

Table 31: Driver Selection Table